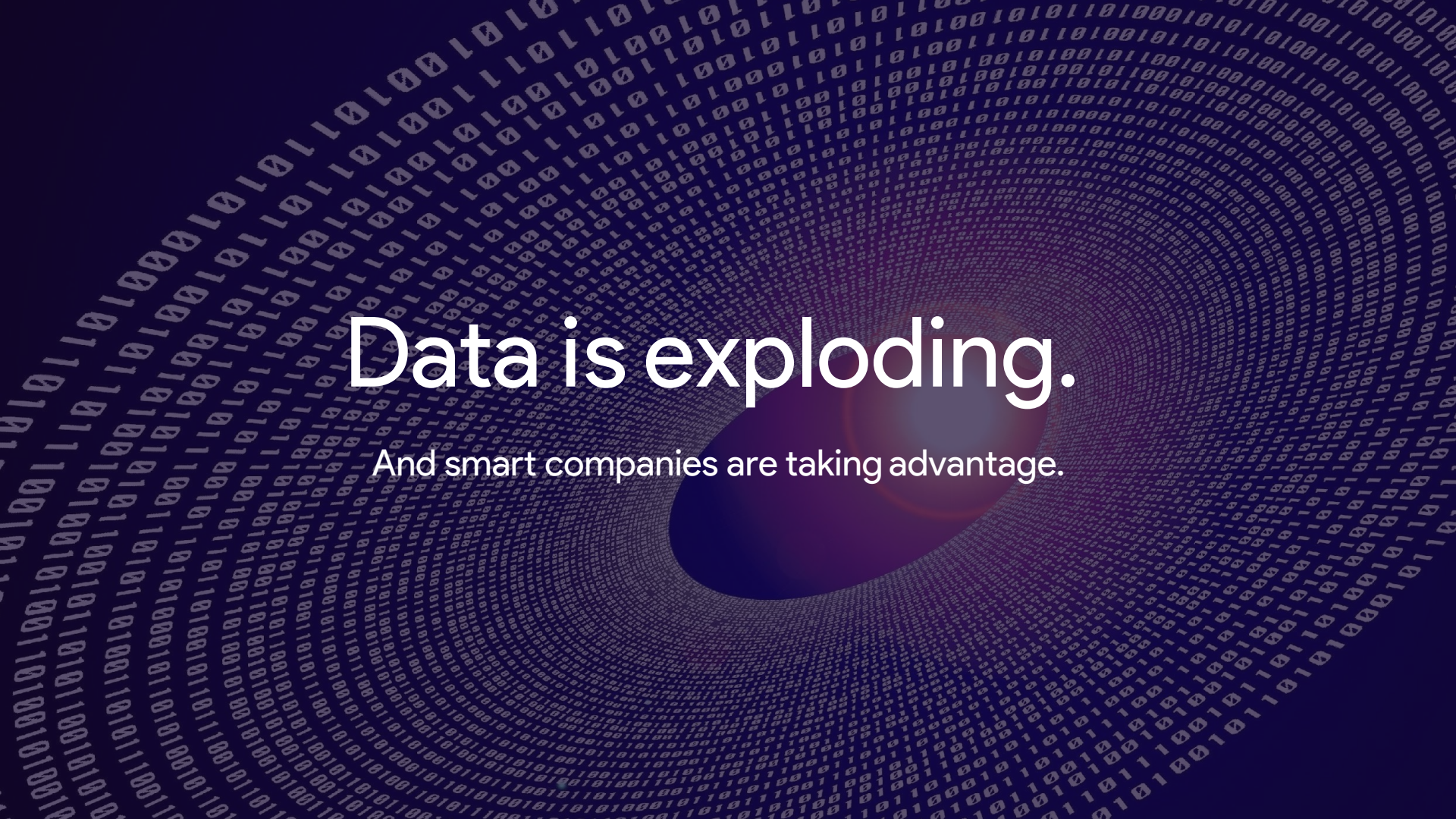
Google Cloud Platform

Cloud Machine Learning

Marc Anthony Reyes (@marcreyesph)


Data is exploding.
And smart companies are taking advantage.

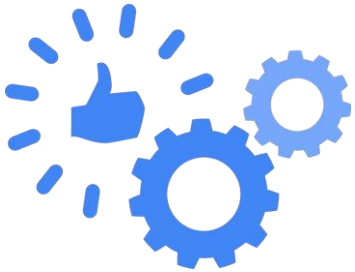
Unstructured data accounts for **90%** of enterprise data*

Cloud Machine Learning help you **make sense of it**

*Source: IDC

What is Machine Learning?

Data



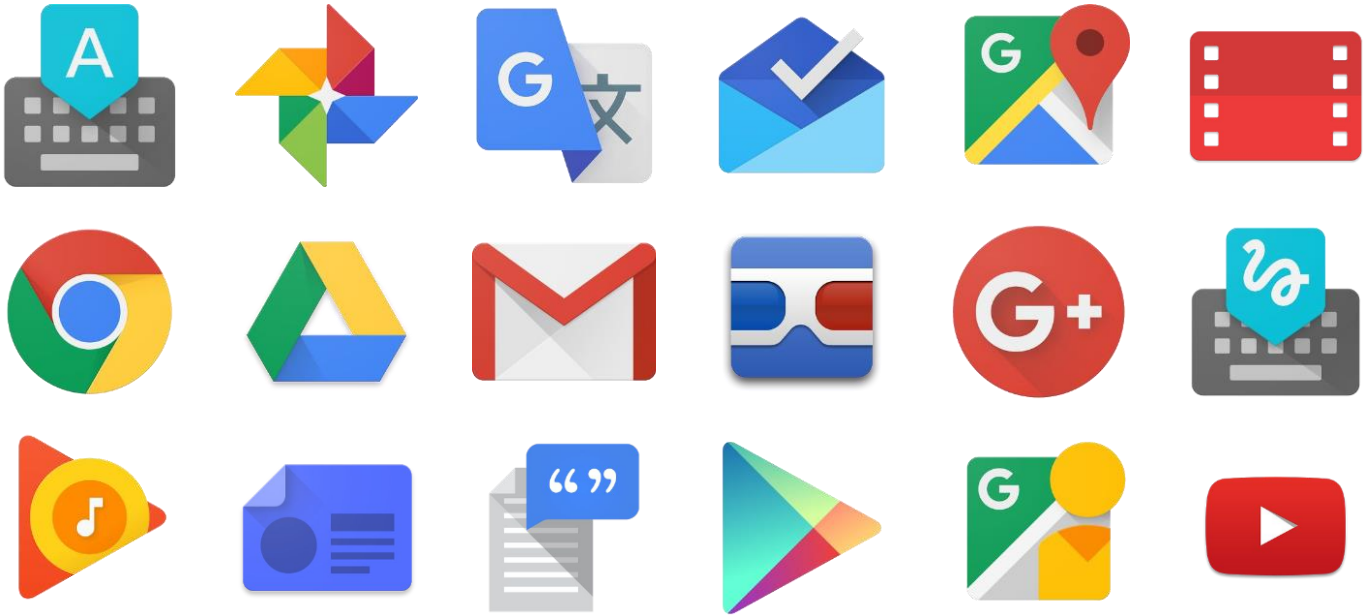
Algorithm

Insight

Machine Learning @Google

Coast
Water
Pool
Beach
Woman
EXCHANGE
INFORMATIO
ACCOMMODAT
EXCURSIO
OUVENIRS / P

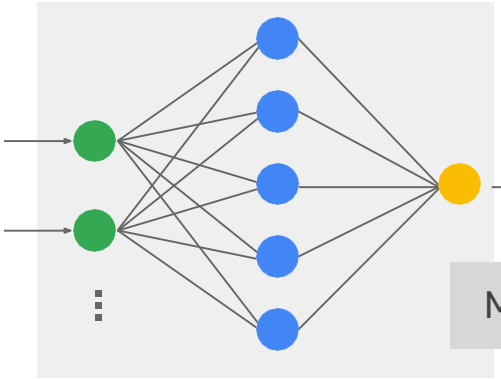
Google Translate



The point of ML is to make predictions

	mintemp	maxtemp	rain	day_1	day_2	day_3	day_4	day_5	day_6	day_7
104	28.9	37.9	0.01	0	0	0	1	0	0	0
9	32.0	43.0	0.00	0	1	0	0	0	0	0
114	35.1	48.0	0.00	1	0	0	0	0	0	0
11	30.0	37.9	0.00	0	0	0	0			
316	19.0	39.9	0.05	0	0	1	0	0	0	0

Input Feature



Predicted Value

Tensorflow helps you “train” models

	mintemp	maxtemp	rain	day_1	day_2	day_3	day_4	day_5	day_6	day_7
104	28.9	37.9	0.01	0	0	0	1	0	0	0
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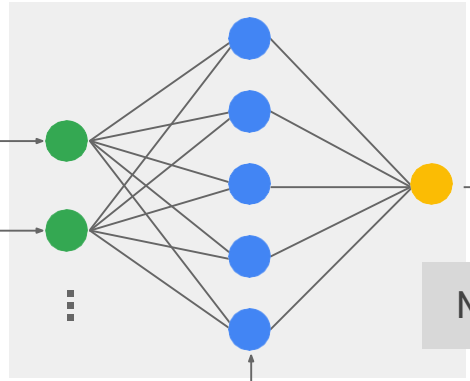


Model

Predicted Value

Cost

Update model based on Cost

True Value

Democratizing machine learning

TensorFlow

CloudML

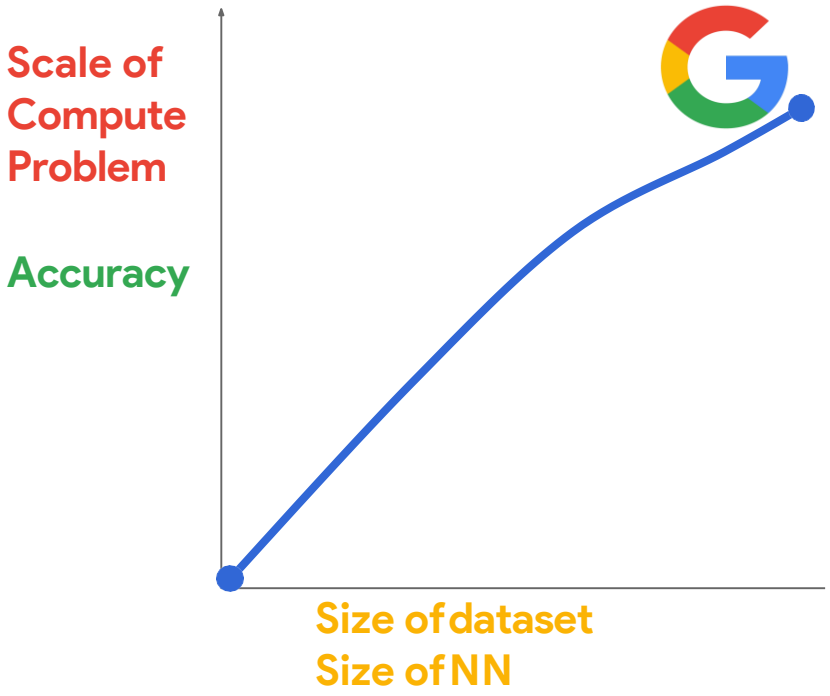
Scale, No-ops
Infrastructure

ML APIs

- Translate API
- Vision API
- Speech API
- Language API

Use/extend OSS SDK	Build custom models	Use pre-built models
ML researcher	Data Scientist	App Developer

Beyond Tensorflow



CloudML (a)

Deep networks

TensorFlow Processing Units (TPUs)

Distributed

No-ops

ML APIs

Translate API

Vision API

Speech API

Language API

https://cloudplatform.googleblog.com/2016/05/Google-supercharges-machine-learning-tasks-with-custom-chip.html

ML APIs are simply REST calls and can be made from any language or framework

```
sservice = build('speech', 'v1beta1', developerKey=APIKEY)
response = sservice.speech().syncrecognize(
    body={
        'config': {
            'encoding': 'LINEAR16',
            'sampleRate': 16000
        },
        'audio': {
            'uri': 'gs://cloud-training-demos/vision/audio.raw'
            }
        }).execute()
print response
```

Data on Cloud Storage

Enterprise Predictive Analytics Challenges

Data access to avariety of data sources.

Data preparation, exploration and visualization.

Develop and build analytic models.

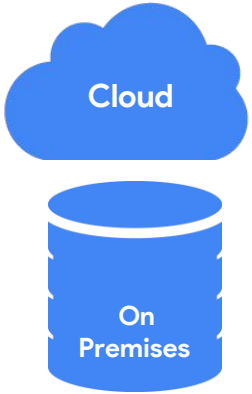
Deploy models and integrate them into businessprocesses and applications.

Perform platform, project and model management.

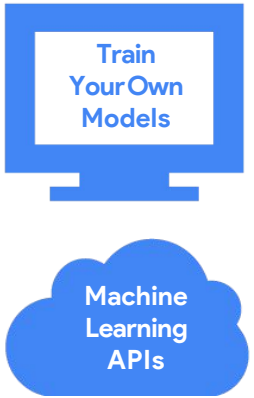
High performance and scalability for both development and deployment.

Data Warehouse is the Foundation of Something Bigger

Data Warehouses/Lakes



Machine Intelligence



Predictive

+

Prescriptive Analytics

=

Advanced Analytics

Machine Learning Use Cases

Manufacturing

- Predictive maintenance or condition monitoring
- Warranty reserve estimation
- Propensity to buy
- Demand forecasting
- Process optimization

Retail

- Predictive inventory planning
- Recommendation engines
- Upsell and cross-channel marketing
- Market segmentation and targeting
- Customer ROI and lifetime value

Healthcare and Life Sciences

- Alerts and diagnostics from real-time patient data
- Disease identification and risk satisfaction
- Patient triage optimization
- Proactive health management
- Healthcare provider sentiment analysis

Travel and Hospitality

- Aircraft scheduling
- Dynamic pricing
- Social media – consumer feedback and interaction analysis
- Customer complaint resolution
- Traffic patterns and congestion management

Financial Services

- Risk analytics and regulation
- Customer Segmentation
- Cross-selling and up-selling
- Sales and marketing campaign management
- Credit worthiness evaluation

Energy, Feedstock and Utilities

- Power usage analytics
- Seismic data processing
- Carbon emissions and trading
- Customer-specific pricing
- Smart grid management
- Energy demand and supply optimization

Why So Little Machine Learning Apps Out There?

- Building and scaling machine learning infrastructure is hard
- Operating production ML system is time consuming and expensive

Building Smart Applications Today

Technology	Operationalization	Tooling
Difficult to scale	Complex data pipelines	Complex dev pipeline with many combinations of tools/libraries
Many choices for different use cases	Managing ML infra takes away time from actually doing ML	Not fully interactive developer experience - collaboration/sharing is hard
Using latest technology (e.g. DNN) is hard	Many models to manage	

Introducing Cloud Machine Learning

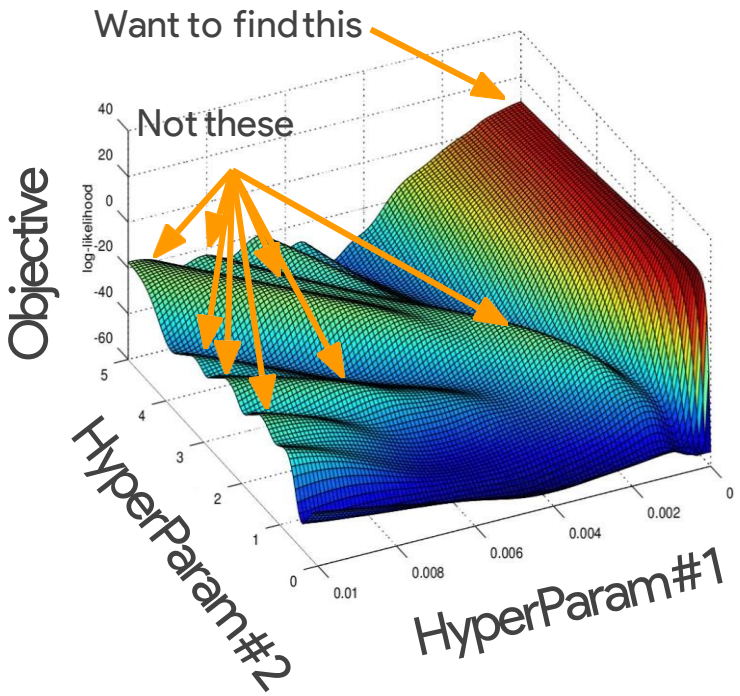
- Fully managed service
- Train using a custom TensorFlow graph for any ML usecases
- Training at scale to shorten dev cycle
- Automatically maximize predictive accuracy with HyperTune
- Batch and online predictions, at scale
- Integrated Datalab experience

Powerful Machine Learning Algorithm

- Convolutional Neural Network for image classification
- Recursive Neural network for text sentiment analysis
- Linear regression at scale to predict consumer action (purchase prediction, churn analysis)
- And unlimited variety of algorithms you can build using TensorFlow

Automatically tune your model with HyperTune

- Automatic hyperparameter tuning service
- Build better performing models faster and save many hours of manual tuning
- Google-developed search algorithm efficiently finds better hyperparameters for your model/dataset



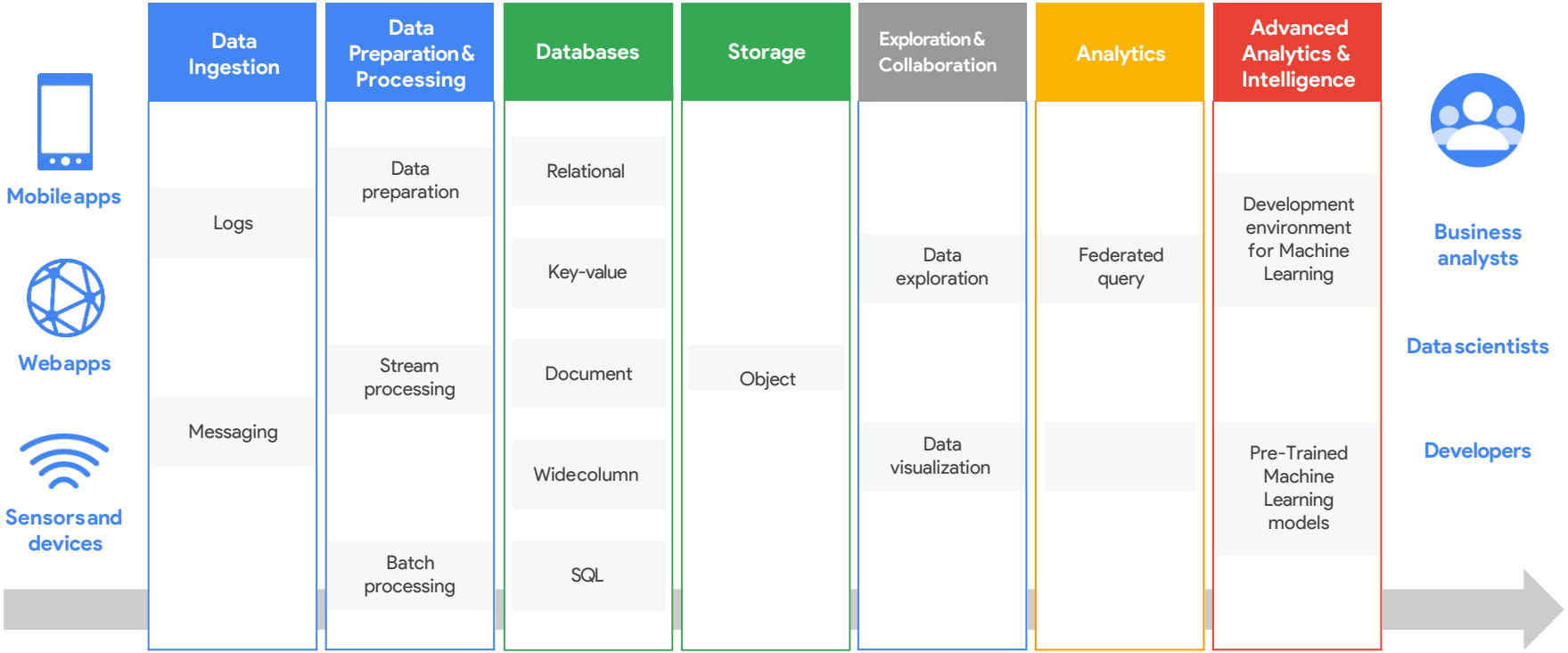
Integrated with GCP Products

- Access data that is stored in GCS or BigQuery
- Save trained models to GCS
- Preprocess largest datasets (TB) using Dataflow
- Orchestrate ML workflow as a Dataflow pipeline
- Analyze data and interactively develop ML models in Datalab
- AutoML for premade and customized ML models curated for your specific ML projects

Fully Managed Machine Learning Services

- Scalable and distributed training infrastructure for your largest data sets
- Scalable prediction infrastructure that can serve very large traffic
- Managed no-ops infrastructure handles provisioning, scaling, and monitoring so that you can focus on building your models instead of handling clusters

Transform Data into Actions



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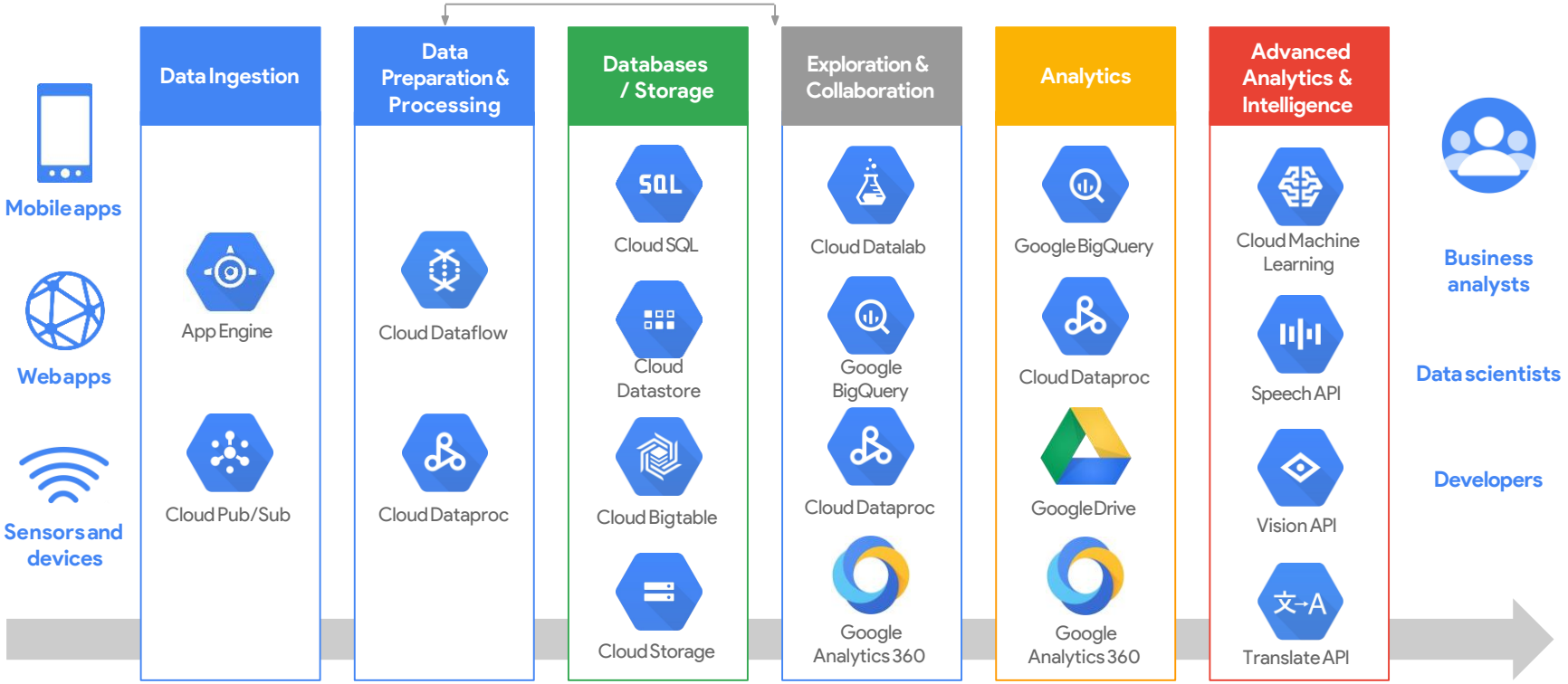
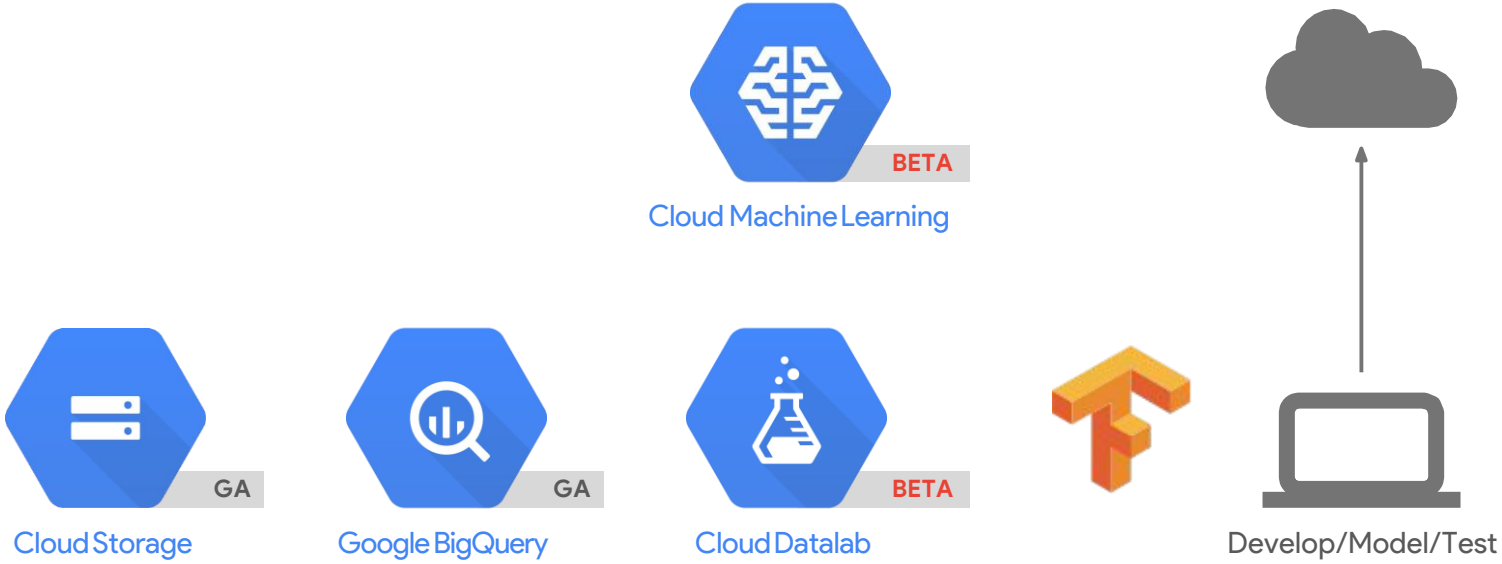


Use Your Own Data to Train Models



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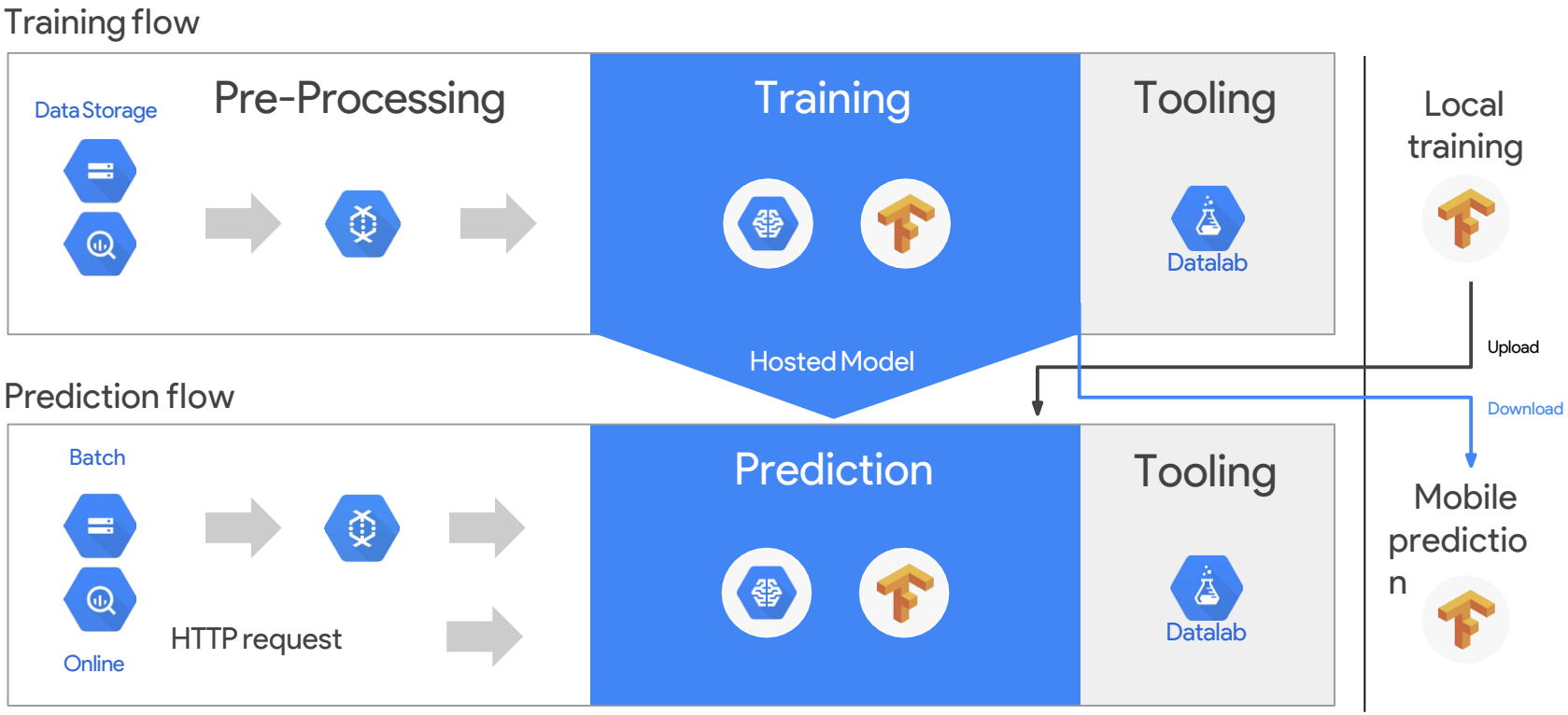
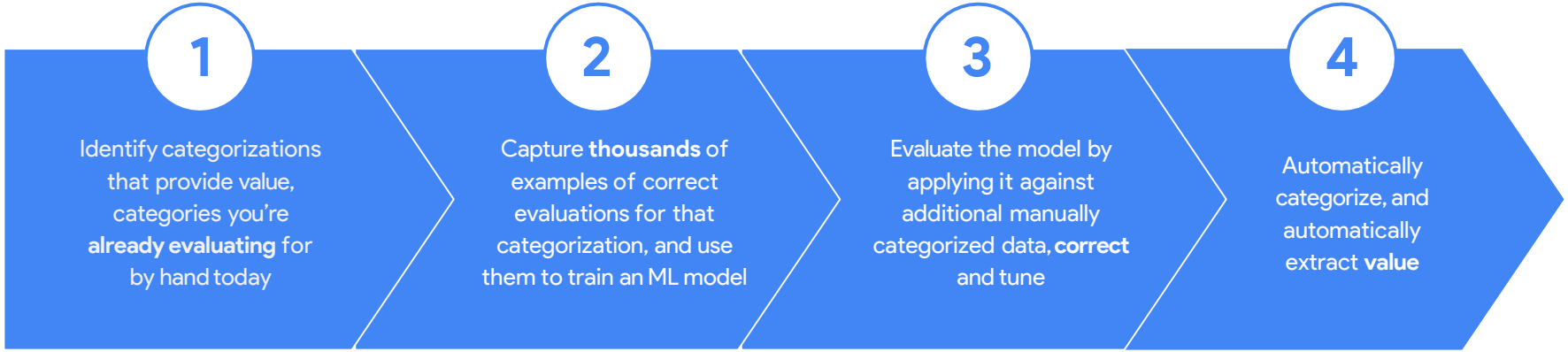


Machine Intelligence is Already Making a Huge Difference and There are Many, Many More Opportunities



Introduction to GCP Cloud ML with Qwiklabs

Codelabs this Afternoon

- Cloud ML Engine: Qwik Start
- Cloud Natural Language API: Qwik Start

Awesome Stuff for Participants

- One-month free access to Qwiklabs with 150 credits for free platform usage.
- One-month free access to GCP Introductory Course on Coursera.

Access These Links for the Qwiklab Activity

- Baseline: Data, ML, AI Quest
- Cloud ML Engine: Qwik Start (http://bit.ly/GCP18CDOCloudML)
- Cloud Natural Language API: Qwik Start (http://bit.ly/GCP18CDONL)
- This afternoon, we only will be taking Cloud ML Engine: Qwik Start and Cloud Natural Language API: Qwik Start

Let’s get our hands dirty with code

cloud.google.com